Case 1: Helmet grades and road exposure

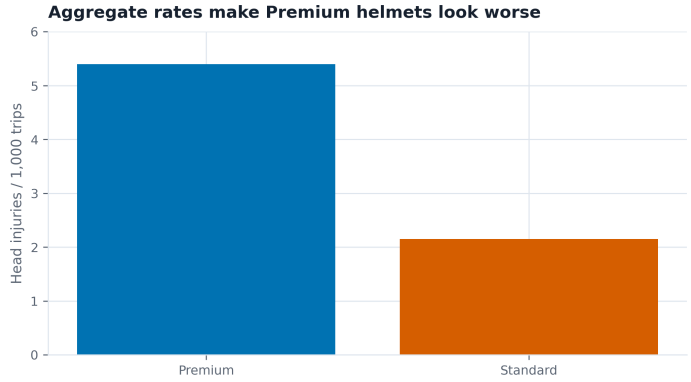
Conclusion: The aggregate chart invites a claim that Premium certification is less safe.

Decoy: A truncated baseline cannot explain this: the bars start at zero. Speed alone is not the mechanism: within-road mean speeds differ only about 1-2 km/h, and Premium rates remain lower within every road type.

Mechanism: Simpson's paradox from road-type composition: 85% of Premium trips but only 3.33% of Standard trips are motorway trips. Motorway rates are 6 vs 8, arterial 2.2 vs 3, lanes 1 vs 1.395 (Premium vs Standard). Common road weights give {'Premium': 3.1, 'Standard': 4.173947368421052} injuries/1,000.

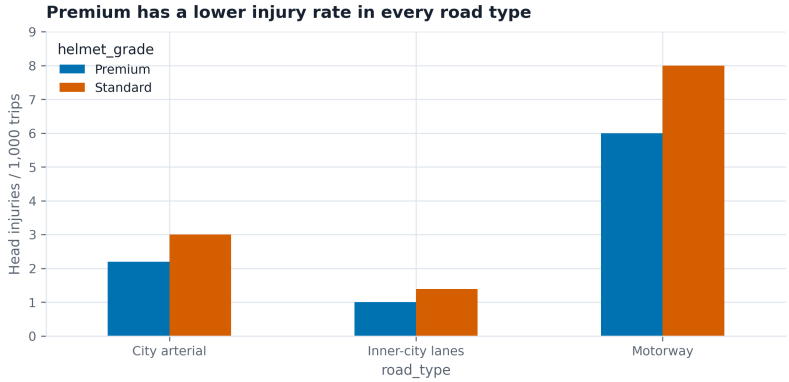
Redesign: Compare trip-weighted rates within road type, with shared zero baseline; no causal helmet-effect estimate is claimed.

Required reconstruction (or summary table)



Source: synthetic CDB601220 Week 3 course data; not real-world evidence.
n=1,000 riders; trip-weighted rates; no rows removed; zero baseline.

Honest redesign



Source: synthetic CDB601220 Week 3 course data; not real-world evidence.
All riders retained; road-specific trip denominators; zero baseline.

Case 2: Counts share an enrolment denominator

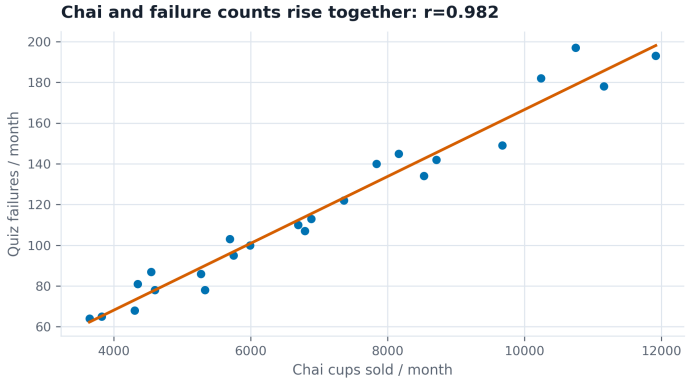
Conclusion: The published chart suggests chai consumption predicts failure.

Decoy: Exam timing is a plausible first objection, but it cannot explain the dominant count trend: enrolment correlates 0.997 with cups and 0.977 with failures; exam status correlates only 0.035 with enrolment.

Mechanism: Population-size confounding: raw r=0.982, per-student r=0.293. Exam-month mean failure rates are 3.862% vs 3.415% in teaching months, so a residual association should not be erased or called zero.

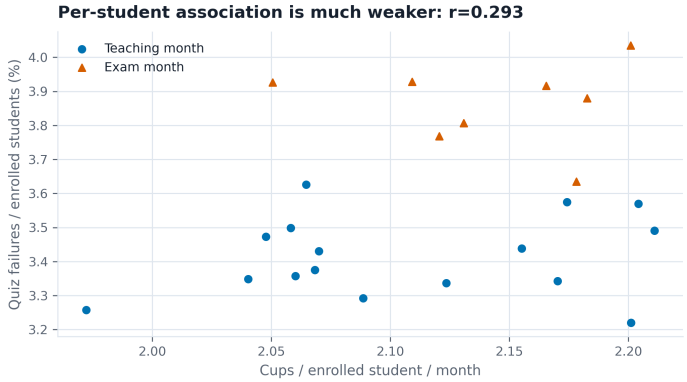
Redesign: Use per-student monthly rates and separate exam months by colour and shape; these are failure events, not a proven individual causal effect.

Required reconstruction (or summary table)



Source: synthetic CDB601220 Week 3 course data; not real-world evidence.
n=24 months; ordinary least squares; all rows retained.

Honest redesign



Source: synthetic CDB601220 Week 3 course data; not real-world evidence.
Denominator: students_enrolled; no rows removed; markers also identify exam status.

Case 3: Missing monitors change the average

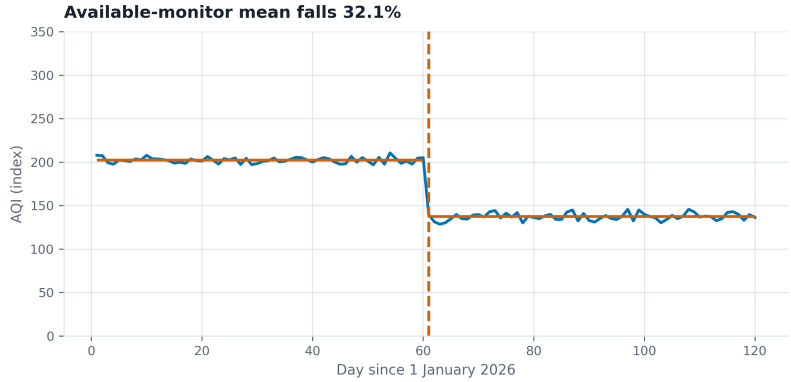
Conclusion: The chart invites attributing a 32% AQI improvement to the action plan.

Decoy: The baseline is zero, so truncation is ruled out. A citywide seasonal improvement is not supported by the continuously reporting panel.

Mechanism: Informative missingness / changing network composition: 6 industrial and 3 arterial stations stop reporting; 540 AQI blanks, reporting stations fall 22 to 13. Available means 202.09 to 137.21; balanced means 137.09 to 137.21, change +0.08%.

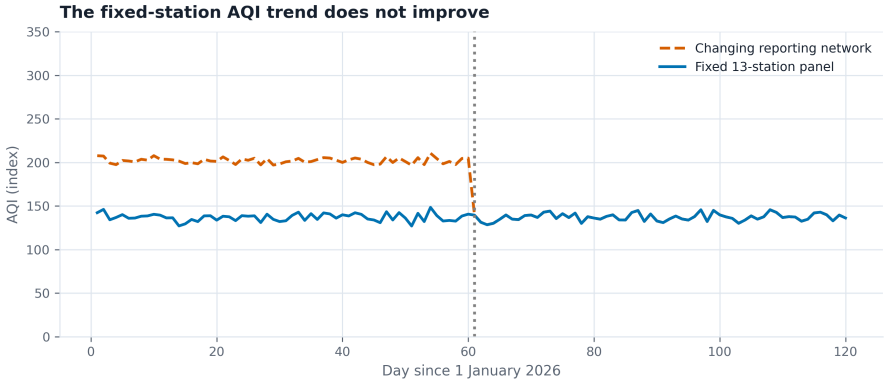
Redesign: Show raw and fixed-station trends together; do not fill missing high-pollution stations with zeros or assert full-city improvement.

Required reconstruction (or summary table)



Source: synthetic CDB601220 Week 3 course data; not real-world evidence.
2,640 station-day rows; 540 offline blanks skipped by mean, not imputed as zero.

Honest redesign



Source: synthetic CDB601220 Week 3 course data; not real-world evidence.
Balanced sensitivity panel retains 1560 rows; excludes 1080 rows from incomplete stations. Raw series remains visible.

Case 4: Equal statistics conceal the quartet

Conclusion: Equal summaries invite equal funding and identical department models.

Decoy: Small n alone is not the explanation: all departments have the same n=11, means approximately 18/14.5 and r approximately .816, yet very different geometry.

Mechanism: Insufficient summaries: one department is roughly linear, one curved, one driven by a score outlier, and Architecture has only two distinct x values (10 at one x, one high-leverage point). Removing that x-extreme leaves constant x, so Pearson r is undefined, not evidence of a stable slope.

Redesign: Plot all four departments on common axes; show fitted lines to reveal where identical fits misrepresent the observations.

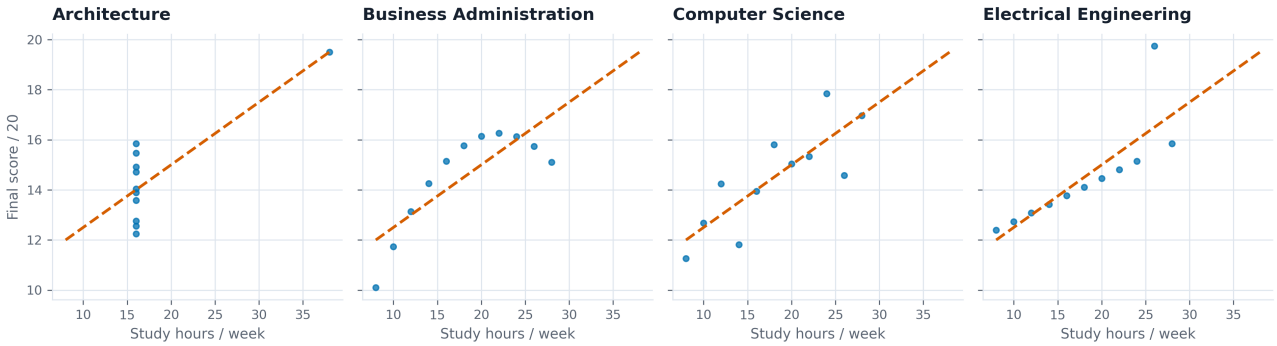
Required reconstruction (or summary table)

Matching summaries hide different geometries

	Architecture	Business Administration	Computer Science	Electrical Engineering
n	11.0	11.0	11.0	11.0
mean_x	18.0	18.0	18.0	18.0
mean_y	14.501	14.501	14.501	14.5
sd_x	6.325	6.325	6.325	6.325
sd_y	1.936	1.937	1.937	1.936
r	0.817	0.816	0.816	0.816
slope	0.25	0.25	0.25	0.25

Source: synthetic CDB601220 Week 3 course data; not real-world evidence.
SD uses ddof=0; slope is ordinary least squares.

Honest redesign



Source: synthetic CDB601220 Week 3 course data; not real-world evidence.
All rows retained; common limits; dashed OLS lines; population SD in summary tables.

Case 5: Moment matching preserves different shapes

Conclusion: Equal means, SDs and correlations invite treating five cohorts as interchangeable.

Decoy: Not a small-sample artifact: each cohort has n=180 and deliberately matched moments. Population rather than sample SD is the convention behind the 12/15 self-check.

Mechanism: Non-identifiability from first/second moments: all five groups have mean 48/63, SD 12/15 and r=.420 to rounding, while scatter geometry differs. Covariance is about 75.6 and slope .525 for all; neither constrains clusters, curves or gaps.

Redesign: Common-scale scatter small multiples expose the shapes without changing any row. Cohort names label the designed shapes, not automatically their final geometry after moment matching.

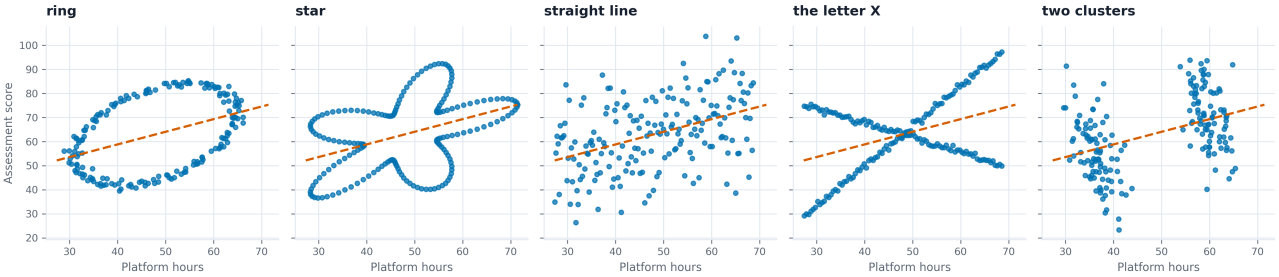
Required reconstruction (or summary table)

Matching summaries hide different geometries

	ring	star	straight line	the letter X	two clusters
n	180.0	180.0	180.0	180.0	180.0
mean_x	48.0	48.0	48.0	48.0	48.0
mean_y	63.0	63.0	63.0	63.0	63.0
sd_x	12.0	12.0	12.0	12.0	12.0
sd_y	15.0	15.0	15.0	15.0	15.0
r	0.42	0.42	0.42	0.42	0.42
slope	0.525	0.525	0.525	0.525	0.525

Source: synthetic CDB601220 Week 3 course data; not real-world evidence.
SD uses ddof=0; slope is ordinary least squares.

Honest redesign



Source: synthetic CDB601220 Week 3 course data; not real-world evidence.
All rows retained; common limits; dashed OLS lines; population SD in summary tables.

Case 6: An influential point and redundant channels

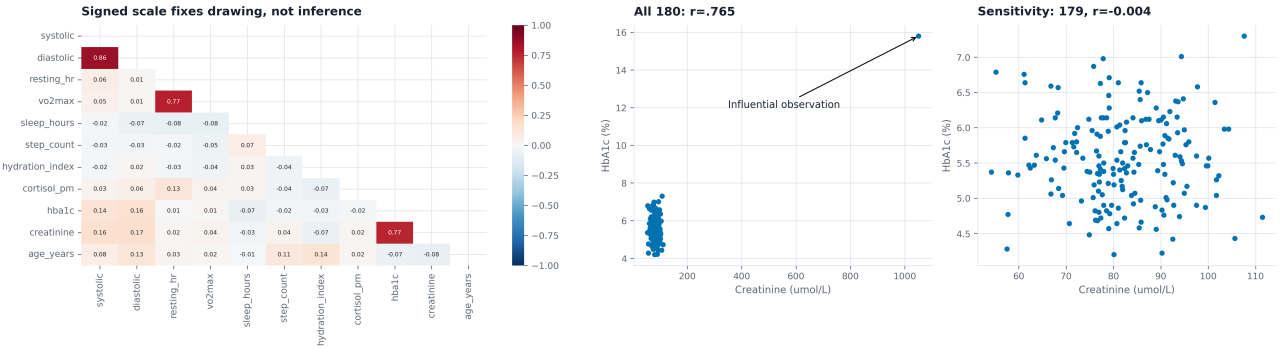
Conclusion: The heatmap invites a new physiological link between creatinine and HbA1c.

Decoy: The positive association is not mainly site confounding: the row at index 0 drives it even within Site C. An extreme value is not automatically erroneous and is retained in the primary analysis.

Mechanism: Outlier influence: r=0.765, versus -0.0036 without the 1050/15.8 observation. Other strong pairs include arithmetic/redundant channels (systolic-MAP .962). Resting-HR/VO2max has pooled r=.772 but within-device r={'DX-100': -0.41, 'DX-200': -0.404, 'DX-300': -0.495, 'DX-400': -0.631}, showing device/site mixing; probes have no calibrated interpretation. With 20 variables there are 190 distinct pairwise tests, requiring exploratory caution.

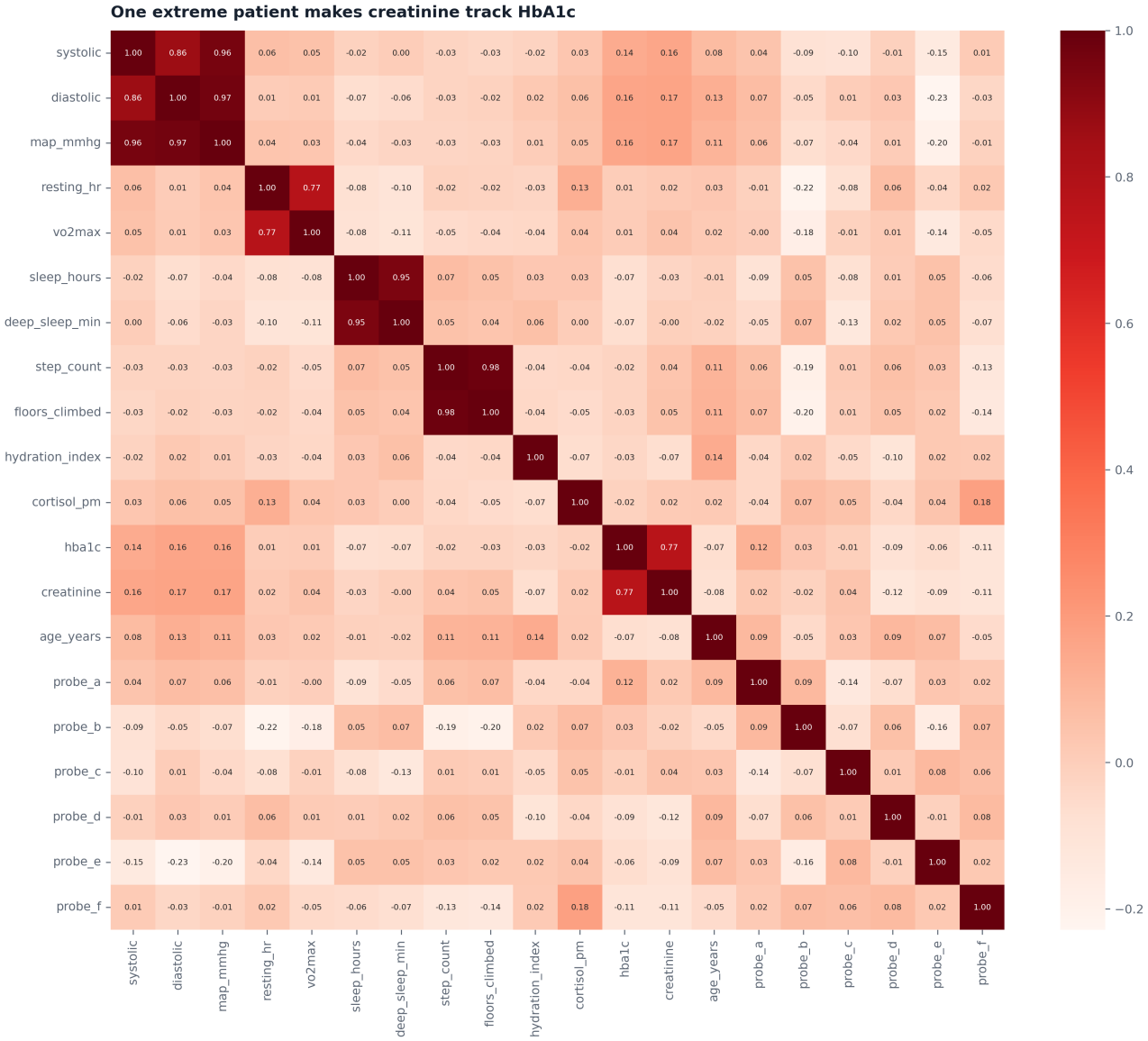
Redesign: A signed -1..1 diverging lower-triangle matrix repairs sign/scale/duplication. It cannot repair leverage, confounding or multiplicity: add raw and leave-one-out scatter, with exclusions explicit; omit uncalibrated and redundant channels from interpretation.

Redesign and disclosed influence sensitivity



Source: synthetic CDB601220 Week 3 course data; not real-world evidence.
Raw heatmap retains 180 rows. Sensitivity excludes exactly row index 0 (1050 umol/L, 15.8%); this is influence analysis, not an asserted device fault. Derived/redundant channels and six uncalibrated probes omitted from redesigned matrix.

Case 6: required original 20 x 20 matrix



Source: synthetic CDB601220 Week 3 course data; not real-world evidence.
Analyst recipe intentionally uses automatic colour limits, Reds and both triangles; n=180.

Case 7: Independent axes manufacture a crossing

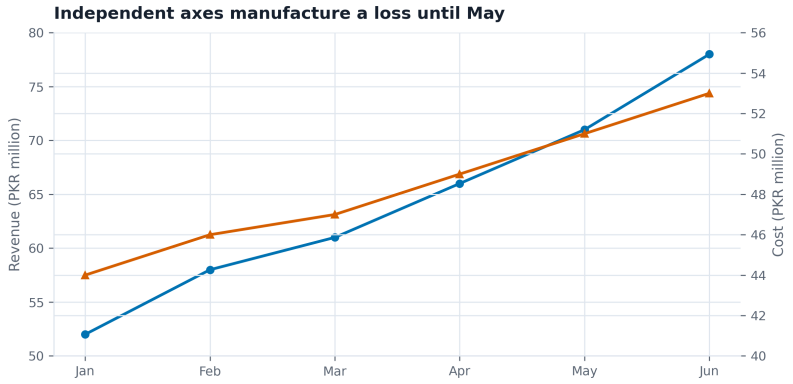
Conclusion: The line positions invite saying the business lost money through April.

Decoy: Not different currencies or arithmetic errors: both columns are PKR millions; actual profits are positive in every row.

Mechanism: Incommensurable normalized positions: January revenue height (52-50)/30=.0667; cost (44-40)/16=.25. Cost is drawn higher even though 44<52. Actual profit grows from 8 to 25 million.

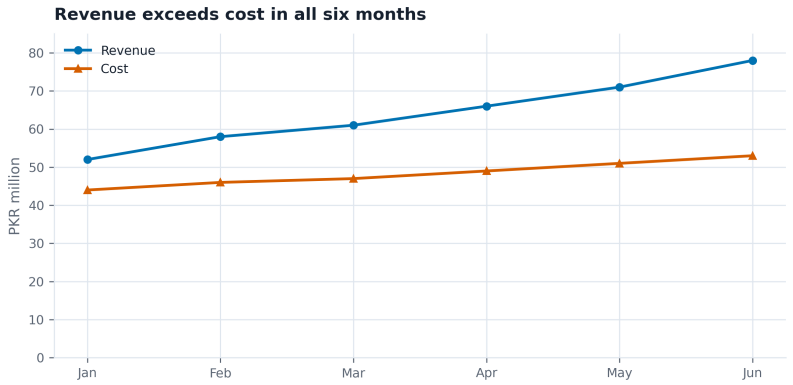
Redesign: Put cost and revenue on one common axis with colour plus different marker shapes.

Required reconstruction (or summary table)



Source: synthetic CDB601220 Week 3 course data; not real-world evidence.
Required reconstruction is the sole dual-axis exception: left 50-80, right 40-56. It is labelled misleading; never used as the redesign.

Honest redesign



Source: synthetic CDB601220 Week 3 course data; not real-world evidence.
Common unit and common zero baseline; monthly profit 8, 12, 14, 17, 20, 25 PKR million.

Case 8: Binning changes the estimator, not the sample

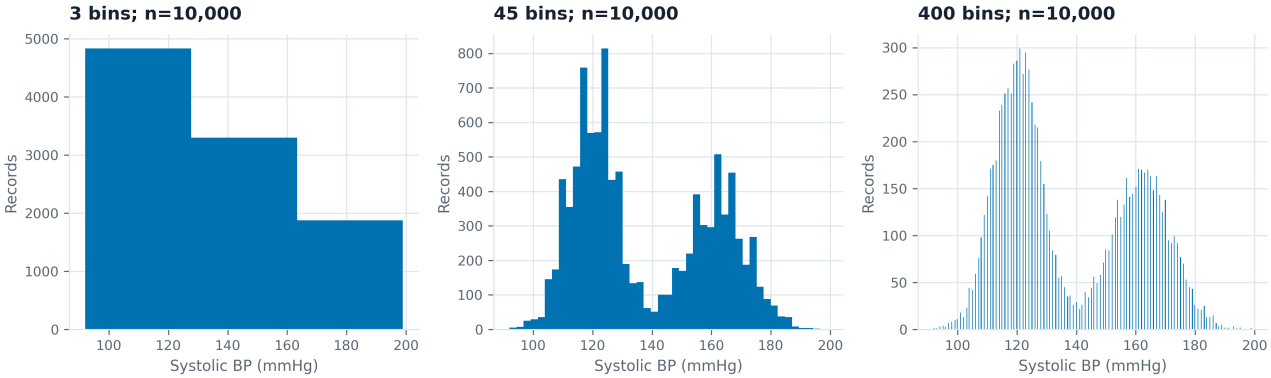
Conclusion: The three panels invite three incompatible distribution stories.

Decoy: Sample-size changes and exclusions are ruled out: n=10,000 with identical min=92 and max=199 in every panel.

Mechanism: Bias/variance and quantization: width 35.67 mmHg (3 bins) merges modes; width .2675 (400 bins) is below the 1 mmHg resolution and generates artificial empty bins. Width 2.378 (45 bins) exposes the mixture with more detail.

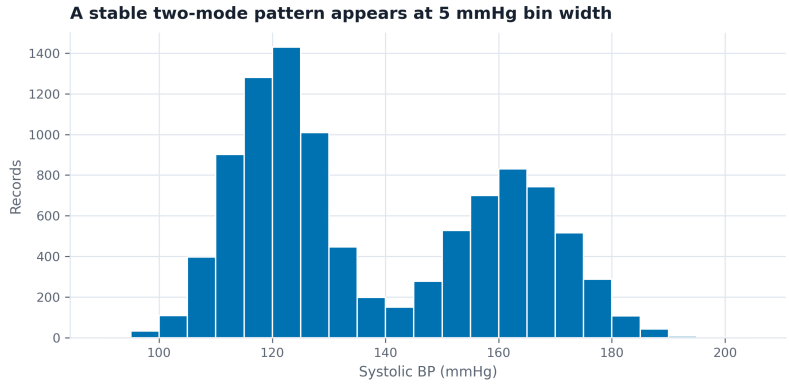
Redesign: Use 5 mmHg equal-width bins, compare all three alternatives, and retain all observations; report two modes rather than claim normality.

Required reconstruction (or summary table)



Source: synthetic CDB601220 Week 3 course data; not real-world evidence.
All three use identical 10,000 values and full 92-199 mmHg range; bin counts 3/45/400.

Honest redesign



Source: synthetic CDB601220 Week 3 course data; not real-world evidence.
Edges 90..205 step 5; n=10,000; no values excluded. Histogram reveals subpopulations but does not identify their cause.

Case 9: Kernel smoothing crosses a physical boundary

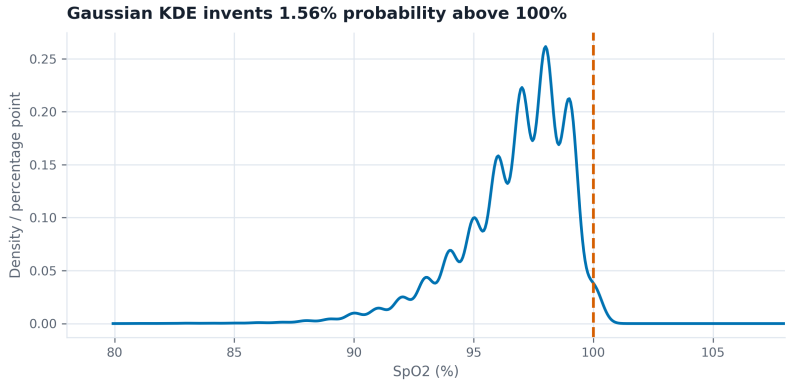
Conclusion: The curve invites believing some patients have saturation above 100%.

Decoy: No impossible source measurements: observed max is exactly 100, with zero rows above it.

Mechanism: Boundary bias from unbounded Gaussian kernels: Scott KDE bandwidth 0.3550 percentage point; integrated mass above 100 is 1.565%. The supplied 4.4% headline does not match this dataset under the specified default Scott bandwidth; the computed result is reported rather than fabricated. Simply setting cut=0 or clip=(0,100) hides the tail but does not correct the estimator or restore its integral.

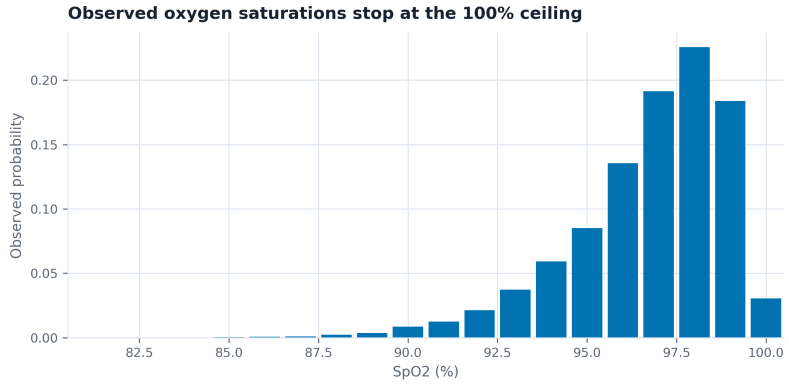
Redesign: Plot empirical discrete probabilities on the bounded integer support; a truncated display of an ordinary Gaussian KDE is not a boundary correction.

Required reconstruction (or summary table)



Source: synthetic CDB601220 Week 3 course data; not real-world evidence.
Scott bandwidth; bw_adjust=1; Gaussian kernel; grid=512; explicit support extension to 108; integral above 100 computed over [100,infinity).

Honest redesign



Source: synthetic CDB601220 Week 3 course data; not real-world evidence.
Discrete empirical probability mass, width .85 percentage point; no smoothing, no exclusions; sum=1.